

Krishna Parvataneni

Cambridge, MA | kveni@mit.edu | [Google Scholar](#)

Education

Massachusetts Institute of Technology (MIT) Master of Engineering in Computer Science, concentration in Bio-EECS	Jun 2026 – May 2027 Cambridge, MA
Massachusetts Institute of Technology (MIT) Bachelor of Science in Artificial Intelligence and Decision-Making	Sep 2023 – May 2026 Cambridge, MA

Awards and Honors

Outstanding UROP Student Award • MIT School of Engineering	May 2026
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Experience

Intuitive Surgical <i>Data Engineer Intern</i>	Jun 2023 – Present Sunnyvale, CA
- Architected a retrieval-augmented generation (RAG) pipeline transforming procedural surgical data and physician queries into auto-generated visual explanations and analytical reports, reducing manual reporting time for engineering stakeholders - Developed Infrastructure-as-Code (IaC) solutions using Terraform to provision and manage AWS, Databricks, and Artifactory resources, enabling reproducible, scalable cloud deployments across environments. - Built analytics frameworks to measure downstream product utility and adoption, delivering actionable performance insights that informed roadmap prioritization for AI-enabled healthcare applications. - Contributed to scalable data platforms supporting clinical decision-support workflows serving surgical teams across multiple hospital systems.	
Massachusetts General Hospital <i>Undergraduate Researcher, Precision in Medical Imaging Lab</i>	Dec 2023 – Present Boston, MA
- Build database powered by LLMs (Qwen, LLAMA, Mixtral) to automate abscess drain, deep vein thrombosis, and venous thromboembolism management for MGH interventional radiology, leveraging parallel computing for scalability and efficiency. - Develop machine learning-based risk assessment models for predicting 30-day rehospitalization of hemodialysis patients following IR procedures.	
MIT Department of Mechanical Engineering <i>Undergraduate Researcher, Traverso Lab</i>	Jul 2023 – Present Cambridge, MA
- Design soft robotics molds with SOLIDWORKS/Fusion360. - Design and test (using ELISA) a star device for oxalate oxidase delivery to treat hyperoxaluria. - Design bio-inspired needles using generative AI techniques.	
MIT McGovern Institute <i>Undergraduate Researcher, Graybiel Lab</i>	Jul 2023 – Jan 2026 Cambridge, MA
- Employed data analysis techniques and machine learning to analyze neuronal recording signals and movement kinematics during decision-making behaviors in freely moving mice. - Conducted protein expression analysis following virus injection in mouse striatum and substantia nigra.	
MIT-Harvard Health Sciences and Technology <i>Undergraduate Researcher, Edelman Lab</i>	Jul 2023 – Feb 2025 Cambridge, MA
Used GANs and computer vision to classify swine echocardiography data, supporting mechanical pacemaker research.	
Kellis Lab, MIT Computer Science and AI Laboratory <i>Research Intern</i>	Aug 2023 – Feb 2024 Cambridge, MA
Developed a single-cell sequencing pipeline to study structural variants linked to neuropsychiatric disease.	

Teaching Engagements

Introduction to Computer Vision, MIT 6.s058/6.4300

Spring 2026

Teaching Assistant to Bill Freeman, Kaiming He, and Vince Monardo

- Led office hours, facilitated review sessions, answered student questions, and supported course logistics.
- Created comprehensive review guides and led review sessions for both the midterm and final examinations.

Introduction to Probability, MIT 6.3700

Fall 2025

Teaching Assistant to William Oliver

- Led office hours, facilitated review sessions, answered student questions, and supported course logistics.
- Managed and coordinated graders to ensure timely and consistent evaluation of problem sets and exams.

Collaborative Data Science in Healthcare, Harvard T.H. Chan SPH BST.209

Summer 2023, 2024

Teaching Assistant to Leo A. Celi, Tom Pollard, and Heather Mattie

- Assisted in teaching BST.209, a machine learning/data science course in R for doctors.
- Collaborated with a team of 5 doctors to develop a project predicting disposition of patients presenting to the ED with Coronary Heart Failure, with a focus on bias reduction.

Extracurriculars

Harvard Global Alliance for Medical Innovation

Aug 2024 – Present

Co-President, Project Team Lead

Cambridge, MA

- Co-lead organization strategy, project oversight, and interdisciplinary initiatives across global health innovation teams.
- Direct development and deployment of AI-based pelvic and acetabular fracture detection software for clinical use in Malawi, in partnership with Beth Israel Deaconess Medical Center.
- Co-mentor a student team developing cardiac wall motion analysis software to support early diagnosis of cardiovascular disease.

Peer-Reviewed Publications

K Parvataneni, DC Schramm, J Cui, CP Bridge, D Daye. *AI-Enabled Risk Prediction for Sepsis in Hemodialysis Patients Undergoing IR Interventions*. In preparation. 2026

K Parvataneni, FJ Dorfner, CP Bridge, GM Foltz, M Alhamami, D Daye. *Open-Weight LLMs for Clinically Actionable DVT Extraction From Venous Duplex Ultrasound Reports*. Under Review. 2026

K Parvataneni, FJ Dorfner, M Preziosi, PD Sutphin, GM Foltz, M Alhamami, CP Bridge, D Daye. *Agentic Large Language Models for Patient-Centered Drain Management in Interventional Radiology*. Under Review. 2026.

M. Zhao[#], Z. Kang[#], **K. Parvataneni**, P. Pang, A. Pettinari, N. Fabian, S. Min, J. Jenkins, K. Ishida, A. Hayward, J. Li, J. Lee, R. Langer, G. Traverso. “Next-generation Gastroretentive Platforms for Oral Protein Delivery as Novel Treatment Modality for Metabolic Disorders”, In submission. ([#]: co-first authors)

M. Zhao, Z. Kang, A. Kirtane, **K. Parvataneni**, N. Fabian, J. Bernstock, A. Pettinari, S. Park, J. Jenkins, C Tang, H. Kim, C. Karavasili, A. Hayward, R. Langer, G. Traverso. “Non-viral Delivery of CRISPR-Cas9 Ribonucleoprotein (RNP) Complex for in vivo Tissue-specific Gene Editing”, In submission.

HA Zaki, A Brea, **K Parvataneni**, H Saab, CP Bridge, S Tripathi, D Daye. *Retrieval-augmented language models for patient-centered periprocedural anticoagulation in interventional radiology*. Cardiovasc Intervent Radiol; 2026 April; <https://doi.org/10.1007/s00270-026-04428-0>

S Gao, R Zhu, P Sui, Z Kong, S Aldogom, Y Huang, A Noori, R Shamji, **K Parvataneni**, T Tsiligkaridis, M Zitnik. *Democratizing AI scientists using ToolUniverse*. 2025 Oct; arXiv:2509.23426

C Vasquez-Venegas, C Wu, S Sundar, R Proa, FJ Beloy, JR Medina, M McNichol, **K Parvataneni**, N Kurtzman, F Mirshawka, M Aguirre-Jerez, DK Ebner, LA Celi. *Detecting and mitigating the clever Hans effect in medical imaging: a scoping review*. J Digit Imaging Inform Med; 2025 Aug; <https://doi.org/10.1007/s10278-024-01335-z>

K Parvataneni, R Tamposi, TC McCloud, MS Gee, JA Brink, D Daye. *Preparing the next generation of healthcare leaders through a longitudinal management and leadership training program for radiology residents*. Curr Probl Diagn Radiol; 2025 April; <https://doi.org/10.1067/j.cpradiol.2025.04.013>

K Parvataneni, R Tamposi, N Mercaldo, MJ Englander, S Josephs, M Uceda, N Monfore, D Daye. *What deters women from pursuing careers in interventional radiology: factors associated with medical trainees decisions to pursue careers in IR*. Curr Probl Diagn Radiol; 2025 Jan; <https://doi.org/10.1067/j.cpradiol.2024.10.023>

I Lazaridis, JR Crittenden, G Ahn, K Hirokane, IR Wickersham, T Yoshida, A Mahar, V Skara, JH Loftus, **K Parvataneni**, K Meletis, JT Ting, E Hueske, A Matsushima, AM Graybiel. *Striosomes control dopamine via dual pathways paralleling canonical basal ganglia circuits*. Curr Biol; 2024 Nov; doi: 10.1016/j.cub.2024.09.070

Y Lin, **K Parvataneni**. *Deepfake Generation, Detection, and Use Cases: A Review Paper*. Int J Comput Biol Intell Syst; 2021 Oct.

Conference Presentations

Abstracts:

K Parvataneni, FJ Dorfner, CP Bridge, D Daye. *Comparative Evaluation of Open-Source Large Language Models for Automated Extraction of Iliofemoral DVT Data from Venous Duplex Ultrasound Reports*. 2026 Society for Imaging and Informatics in Medicine.

K Parvataneni, DC Schramm, J Cui, CP Bridge, D Daye. *AI-Enabled Risk Prediction for Sepsis and Septic Shock in Hemodialysis Patients Undergoing Interventional Radiology Interventions*. 2026 Society for Interventional Radiology.

H Zaki, **K Parvataneni**, A Brea, H Saab, S Tripathi, C Bridge, D Daye. *Retrieval-Augmented Language Models for Patient-Centered Periprocedural Anticoagulation in Interventional Radiology*. 2026 Society for Interventional Radiology.

K Parvataneni, S Tripathi, FJ Dorfner, CP Bridge, D Daye. *Reasoning-Based Large Language Model Study for Automated Drain Management in Interventional Radiology*. 2025 Conference on Artificial Intelligence in Medical Imaging.

K Parvataneni, FJ Dorfner, M Preziosi, PD Sutphin, CP Bridge, D Daye. *Large Language Models for Patient-Centered Care: A Proof of Concept Study for a Drain Management and Clinical Follow-Up Database in Interventional Radiology*. 2025 Society for Imaging and Informatics in Medicine

M Fei, S Tripathi, **K Parvataneni**, FJ Dorfner, CP Bridge, D Daye. *Using Open-Source Large Language Models to Extract Labels from Radiology Reports for Machine Learning Applications: A Proof-of-Concept Study*. 2025 Society for Imaging and Informatics in Medicine

K Parvataneni, FJ Dorfner, CP Bridge, D Daye. *Leveraging Large Language Models for Patient-Centered Care: A Proof of Concept Study on a DVT Management Database*. 2025 GEST (Global Embolization Symposium & Technologies)

K Parvataneni, S Ghandour, J Cui, CP Bridge, D Daye. *AI-Enabled Tool for 30-Day Mortality Risk Stratification in Hemodialysis Patients Undergoing Interventional Radiology Procedures*. 2025 Society for Interventional Radiology

K Parvataneni, SH Zaidi. *Comparative Study on Bacterial Inactivation using O₂ and N₂-added He/Ar Plasma*. 2024 International Conference on Plasma Medicine

K Parvataneni, D Yilmaz, A Dawar, S Padmanabhan, S Bajaj, S Bokkisam, V Dinesh, S Jahanikia. *Using Topological Data Analysis (TDA) and Mapper to Study the Homology of Neurodegenerative Diseases*. 2023 Organization for Human Brain Mapping

B Bellannagari, **K Parvataneni**, D Prasun, A Dogariu, SH Zaidi. *Bacterial Inactivation by using a DBD Plasma Sheet Generator*. 2022 International Conference on Plasma Medicine

K Parvataneni, SH Zaidi. *Investigating Rotational and Vibrational Temperatures of a DBD Plasma Jet for Medical Applications*. 2022 Southern California Conference on Undergraduate Research

K Parvataneni, SH Zaidi. *Design Improvements in Dielectric Barrier Discharge Plasma Torches for Medical Applications*. 2022 APS Division of Plasma Physics Meeting

K Parvataneni, A Joseph, F Nomani, J Jadav, D Mundewadi, SH Zaidi. *Arduino-Based Plasma Temperature Measurements in Wound Healing and Sterilization Process*. 2022 Bay Area Biomedical Device Conference

S Tammineedi, **K Parvataneni**, SH Zaidi. *Suitability of Dielectric Barrier Discharge Plasma Torches for Medical Applications*. 2022 Bay Area Biomedical Device Conference

S Tammineedi, **K Parvataneni**. *Automated Temperature Control of a DBD Plasma Jet for Wound Healing and Sterilization*. 2022 National Conference on Undergraduate Research (NCUR)

S Tammineedi, **K Parvataneni**. *Investigating Plasma Luminosity and Temperatures along a DBD Plasma Jet Designed for Medical Applications*. 2021 Southern California Conference on Undergraduate Research Conference

Full Papers:

M Pinnaka, **K Parvataneni**, SH Zaidi. *Impact of Data Heterogeneity on AI/ML Model Accuracy in Assisting Pneumonia Type Prediction*. 2024 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communications Technology

A Tummala, **K Parvataneni**. *Comparative study of AI-based anemia prediction using eye conjunctiva images and blood tests*. 2023 International Workshop on Artificial Intelligence and Image Processing

A Tummala, **K Parvataneni**. *Early prediction of chronic kidney disease using AI*. 2023 15th Biomedical Engineering International Conference

K Parvataneni, F Kazmi, SHA Kazmi, SH Zaidi. *Applying Machine Learning on Breast Mass Digitized Images to Predict Breast Cancer*. 2023 International Conference on Machine Learning and Cybernetics

D Gopakumar, V Nair, **K Parvataneni**. *An Artificial Intelligence Approach to Fetal Health Risk Prediction*. 2023 IEEE Integrated STEM Education Conference

K Parvataneni, SH Zaidi. *Exploring the Vibrational and Rotational Temperatures of a DBD Plasma Jet for Wound Healing*. 2022 IEEE-EMBS Conference on Biomedical Engineering and Sciences

W Xu, J Zhao, **K Parvataneni**, V Zhang, S Lin, H Wang. *Hybrid AI Framework for Remote Patient Heart Failure Risk Prediction*. 2022 Global Reliability and Prognostics and Health Management. **Best Paper Award**.